

Score: / 10

Calculus I (MATH 2200)

Spring 2020

Wednesday, 03/04/2020

Quiz 6: Section 3.5, 3.6, 3.7

Name (Print): _____

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Section 3.5

2 pts

1. Find $\frac{dy}{dx}$ given that $y = \sec(x) \csc(x)$.

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(\sec(x) \csc(x)) \\ &= \frac{d}{dx}(\sec(x))(\csc(x)) + (\sec(x))\frac{d}{dx}(\csc(x)) \\ &= (\sec(x) \tan(x))(\csc(x)) + (\sec(x))(-\csc(x) \cot(x)) \\ &= \frac{1}{\cos(x)} \frac{\sin(x)}{\cos(x)} \frac{1}{\sin(x)} - \frac{1}{\cos(x)} \frac{1}{\sin(x)} \frac{\cos(x)}{\sin(x)} \\ &= \frac{1}{\cos^2(x)} - \frac{1}{\sin^2(x)} \\ &= \sec^2(x) - \csc^2(x).\end{aligned}$$

2 pts

2. Find $\frac{dy}{dx}$ given that $y = \frac{\sin(x) + 1}{\cot(x)}$.

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} \left[\frac{\sin(x) + 1}{\cot(x)} \right] \\&= \frac{\frac{d}{dx} [\sin(x) + 1] \cot(x) - (\sin(x) + 1) \frac{d}{dx} [\cot(x)]}{(\cot(x))^2} \\&= \frac{\cos(x) \cot(x) - (\sin(x) + 1) (-\csc^2(x))}{\cot^2(x)} \\&= \frac{\cos(x) \cot(x) + (\sin(x) + 1) \csc^2(x)}{\cot^2(x)} \\&= [\cos(x) \cot(x) + (\sin(x) + 1) \csc^2(x)] \tan^2(x) \\&= \left(\cos(x) \frac{\cos(x)}{\sin(x)} + (\sin(x) + 1) \frac{1}{\sin^2(x)} \right) \frac{\sin^2(x)}{\cos^2(x)} \\&= \frac{\cos^2(x)}{\sin(x)} \frac{\sin^2(x)}{\cos^2(x)} + (\sin(x) + 1) \frac{1}{\sin^2(x)} \frac{\sin^2(x)}{\cos^2(x)} \\&= \sin(x) + \sec^2(x) (\sin(x) + 1)\end{aligned}$$

Section 3.6

3 pts

1. Suppose a stone is thrown vertically upward with an initial velocity of 64 ft/s from a bridge 96 ft above a river. By Newton's laws of motion, the position of the stone (measured as the height above the river) after t seconds is

$$s(t) = -16t^2 + 64t + 96,$$

where $s = 0$ is the level of the river. Approximately with what velocity will the stone strike the river?

As velocity is the derivative of position, we find the velocity function to be

$$v(t) = s'(t) = -32t + 64.$$

Now, to determine the velocity at which the stone strikes the river, we must first determine when it reaches the river. The stone strikes the river when

$$s(t) = -16t^2 + 64t + 96 = 0.$$

Dividing both sides of the equation by -16 , we obtain

$$t^2 - 4t - 6 = 0.$$

Using the quadratic formula, the solutions are $t \approx 5.16$ or $t \approx -1.16$. Because the stone is thrown at $t = 0$, only positive values of t are of interest; therefore, the relevant root is $t \approx 5.16$. The velocity of the stone (in ft/s) when it strikes the river is approximately

$$v(5.16) = -32(5.16) + 64 = -101.12.$$

Section 3.7

3 pts

1. Find $\frac{d}{dx} \left[(\tan(x) + 10)^{21} \right]$.

First, we find a way to write the function $y = (\tan(x) + 10)^{21}$ as a composition. Setting $y(u) = u^{21}$ and $u = \tan(x) + 10$ will suffice for our purposes. Calculating, $\frac{dy}{du} = 21u^{20}$ and $\frac{du}{dx} = \sec^2(x)$. Hence, we then have that

$$\begin{aligned} \frac{d}{dx} \left[(\tan(x) + 10)^{21} \right] &= \frac{dy}{dx} \\ &= \frac{dy}{du} \frac{du}{dx} \\ &= 21u^{20} \sec^2(x) \\ &= 21(\tan(x) + 10)^{20} \sec^2(x). \end{aligned}$$